SVKM'S NMIMS UNIVERSITY

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING (MPSTME)

Indore Campus, Madhya Pradesh, India

CAPSTONE PROJECT REPORT

MedTrust: A Unified Permissioned Blockchain Framework for Dual-Layer Verification of Prescription Authenticity and Drug Provenance in Pharmaceutical Dispensing

Project Report submitted in partial fulfilment of the requirements
for the degree of **B.Tech** in **Computer Engineering**

Submitted by

Abhishek Soni **(Project Member)**
Arin Jain **(Project Member)**
Neelansh Dosi **(Project Member)**

Under the supervision of

Faculty Mentor
Department of Computer Engineering, MPSTME

Academic Year 2026–27

Certificate

This is to certify that the project entitled **"MedTrust: A Unified Permissioned Blockchain Framework for Dual-Layer Verification of Prescription Authenticity and Drug Provenance in Pharmaceutical Dispensing"** has been carried out by **Abhishek Soni**, **Arin Jain**, and **Neelansh Dosi** under the guidance and supervision of the undersigned and is submitted in partial fulfilment of the requirements for the degree of **B.Tech** in **Computer Engineering** of Mukesh Patel School of Technology Management & Engineering (MPSTME), SVKM's NMIMS University, India.

Project Mentor	Examiner
Head of Department	Department / Examiner

Date: _____ Place: Indore

Declaration

We hereby declare that the work presented in this capstone project report entitled **"MedTrust: A Unified Permissioned Blockchain Framework for Dual-Layer Verification of Prescription Authenticity and Drug Provenance in Pharmaceutical Dispensing"** is the documented work carried out as part of the capstone project. The report is prepared for academic evaluation and presents the architecture, methodology, implementation approach, experiments, analysis, limitations, applications, and future scope of the MedTrust framework.

The report is based on the project research paper supplied with the capstone documentation. Wherever external scholarly work is discussed, the corresponding numbered reference is cited. No patient personally identifiable information is used in the proposed on-chain data model; the project design instead uses cryptographic digests and off-chain protected records as described in the research work.

Name	**Name**	**Name**
Abhishek Soni	Arin Jain	Neelansh Dosi
Signature	Signature	Signature

Acknowledgement

We express our sincere gratitude to our project mentor and the Department of Computer Engineering, Mukesh Patel School of Technology Management & Engineering, SVKM's NMIMS University, for providing the academic direction, review framework, and technical environment required to carry out this capstone project.

We are grateful to the faculty members who reviewed the project during its development and provided feedback on the problem definition, literature survey, system architecture, experimental methodology, and technical documentation. The capstone guidelines emphasize regular mentor interaction, progress reporting, technical presentation, and documentation; these activities helped structure the project into measurable stages of research, design, implementation, and evaluation.

We also acknowledge the researchers and authors whose work forms the literature foundation of this project. Their contributions to blockchain-enabled e-prescription systems, pharmaceutical traceability, healthcare privacy, prescription monitoring, and distributed ledgers helped establish the research gap addressed by MedTrust.

Finally, we thank our families, peers, and colleagues for their encouragement throughout the project. Their support enabled us to maintain continuity during the literature review, system design, implementation, benchmarking, and preparation of this report.

Name	**Roll No.**	**SAP ID**
Abhishek Soni		
Arin Jain		
Neelansh Dosi		

Abstract

The dispensing of prescription medicines is a safety-critical process in which the validity of a medical authorization and the authenticity of the physical medicine should be established together. The supplied research work identifies a *dual-layer verification gap*: conventional e-prescription systems primarily validate clinical authorization, whereas pharmaceutical track-and-trace systems primarily validate physical custody. When these pathways are not cryptographically bound at the dispensing point, a valid prescription can be paired with an illegitimate medicine, or an authentic medicine can be obtained through an invalid, duplicated, or expired prescription.

This report presents **MedTrust**, a permissioned blockchain architecture implemented around Hyperledger Fabric 2.5. The system combines a Prescription Verification Module and a Drug Provenance Module through an Atomic Gateway Contract. The prescription layer verifies an ECDSA signature over a salted SHA-256 digest and enforces an explicit prescription lifecycle. The drug layer validates package genesis, expiration, recall status, and the recorded custodian. A transaction is committed only when both verification tracks succeed concurrently.

Privacy is incorporated by design. The proposed on-chain state contains no cleartext patient personally identifiable information; instead, a 128-bit cryptographic salt is incorporated into a SHA-256 digest, while sensitive cleartext information is retained in an encrypted off-chain store. The experimental framework described in the research paper uses six organizational tiers, CouchDB world-state storage, MongoDB for protected off-chain information, a Node.js verification gateway, and React-based pharmacy terminals.

Across eight pharmaceutical workload categories and 5,230 simulated dispensing cycles, the reported evaluation achieved 100.0% deterministic interception of the tested adversarial categories, a mean end-to-end latency of 2.44 ms, peak throughput of 1,690 transactions per second under 500 concurrent terminal threads, and a 96.2% reduction in ledger storage compared with the monolithic on-chain baseline described in the study. The report also examines the threat model, system architecture, cryptographic protocol, smart-contract coordination, comparative evaluation, privacy considerations, advantages, limitations, applications, and future extensions.

Keywords: Permissioned Blockchain, Pharmaceutical Traceability, E-Prescription, Hyperledger Fabric, Atomic Verification, ECDSA, SHA-256, DPDP Act 2023, Zero-PII

Ledger.

Contents

Certificate **1**

Declaration **2**

Acknowledgement **3**

Abstract **4**

Abbreviations **viii**

1 Introduction **1**

1.1 Introduction to the Project 1
1.2 Background of the Project Topic 1
1.3 Motivation 2
1.4 Scope of the Project 3
1.5 Problem Statement 3
1.6 Formal Statement of the Verification Gap 4
1.7 Objectives 4
1.8 Salient Contributions 5
1.9 Organization of the Report 5

2 Literature Survey **6**

2.1 Introduction 6
2.2 Distributed E-Prescription Architectures 6
2.3 Blockchain-Based Pharmaceutical Traceability 7
2.4 Healthcare Privacy and Zero-Knowledge Approaches 8
2.5 Centralized Prescription Monitoring 8
2.6 Systematic Comparative Review 8
2.7 Research Gap Identification 9
2.8 Research Gap to Problem Definition 9
2.9 Literature Review Matrix 10

3 Methodology and Implementation **12**

3.1 Introduction 12

3.2 Design Requirements . . . 12
3.2.1 Functional Requirements . . . 12
3.2.2 Non-Functional Requirements . . . 13
3.3 High-Level System Architecture . . . 13
3.4 Consortium Stakeholder Hierarchy . . . 13
3.4.1 Prescribing Physician . . . 14
3.4.2 Patient Credential Wallet . . . 14
3.4.3 Pharmaceutical Manufacturer . . . 14
3.4.4 Logistics Distributor . . . 14
3.4.5 Retail Pharmacy . . . 14
3.4.6 Drug Regulatory Authority . . . 14
3.5 Identity and Membership . . . 14
3.6 Privacy-Preserving Data Model . . . 15
3.7 Prescription Verification Module . . . 15
3.8 Drug Provenance Module . . . 16
3.9 Atomic Gateway Contract . . . 16
3.10 Transaction Workflow . . . 17
3.11 Threat Model . . . 17
3.11.1 Replay and Doctor-Shopping Fraud . . . 18
3.11.2 Counterfeit Packaging Injection . . . 18
3.11.3 Logistics Custody Hijacking . . . 18
3.11.4 Recall Evasion . . . 18
3.11.5 Expired Prescription Redemption . . . 18
3.11.6 Patient Re-identification . . . 18
3.12 Representative Chaincode Logic . . . 18
3.13 Implementation Stack . . . 19
3.14 Testbed Configuration . . . 20
3.15 Methodological Sequence . . . 20

4 Results and Analysis **21**
4.1 Introduction . . . 21
4.2 Workload Design . . . 21
4.3 Latency Decomposition . . . 22
4.4 Concurrency and Throughput . . . 23
4.5 Throughput Plot . . . 24
4.6 Ledger Storage Optimization . . . 24
4.7 Security Interception . . . 25
4.8 Cryptographic Security Analysis . . . 26
4.8.1 Digest Integrity . . . 26

4.8.2 Signature Unforgeability 26
4.8.3 Atomicity 26
4.9 Comparative Evaluation 26
4.10 Interpretation of Results 27
4.11 Result Validation Considerations 27

5 Advantages, Limitations and Applications 29
5.1 Introduction 29
5.2 Advantages 29
5.2.1 Synchronous Dual-Layer Verification 29
5.2.2 Atomic State Transition 29
5.2.3 Privacy-Aware Ledger Design 29
5.2.4 Permissioned Governance 29
5.2.5 Auditability 30
5.2.6 Fast Point-of-Sale Verification 30
5.2.7 Scalability in the Reported Benchmark 30
5.2.8 Storage Efficiency 30
5.3 Limitations 30
5.3.1 Prototype-Level Evaluation 30
5.3.2 Single-Workstation Testbed 30
5.3.3 Dependence on Data Quality 30
5.3.4 Identity and Key Management 31
5.3.5 Integration Complexity 31
5.3.6 Privacy Is Not Equivalent to Complete Anonymity 31
5.3.7 Regulatory Interpretation 31
5.4 Applications 31
5.4.1 Retail Pharmacy Dispensing 31
5.4.2 Hospital Pharmacy 31
5.4.3 High-Risk or Controlled Medicines 31
5.4.4 Cold-Chain and Specialized Medicines 32
5.4.5 Regulatory Recall Enforcement 32
5.4.6 Pharmaceutical Distribution Networks 32
5.5 Ethical and Social Considerations 32
5.6 Security and Privacy by Design 32
5.7 Operational Failure Scenarios 33

6 Conclusion and Future Scope 34
6.1 Conclusion 34
6.2 Future Scope 34

6.2.1 Multi-Host Enterprise Testnet . . . 35
6.2.2 Fabric Gateway SDK Integration . . . 35
6.2.3 Physical Barcode and RFID Hardware . . . 35
6.2.4 Interoperability with ABDM . . . 35
6.2.5 Sensor-Assisted Cold-Chain Validation . . . 35
6.2.6 Key Rotation and Revocation . . . 35
6.2.7 Resilience and Disaster Recovery . . . 35
6.2.8 Broader Adversarial Evaluation . . . 36
6.3 Final Remarks . . . 36

References **37**

A Soft Code Flowcharts **40**
A.1 Prescription Registration Flow . . . 40
A.2 Drug Asset Registration Flow . . . 41
A.3 Atomic Dispensing Flow . . . 41

B Data Structures and Sample Records **42**
B.1 Prescription Ledger Record . . . 42
B.2 Drug Asset Record . . . 42
B.3 Incident Event Record . . . 43

C Experimental Observations **44**
C.1 Workload Totals . . . 44
C.2 Latency Observations . . . 44
C.3 Security Interception Observations . . . 45

D System Components and Specifications **46**
D.1 Computing Testbed . . . 46
D.2 Cryptographic Components . . . 46

E Project Work Plan and Review Documentation **47**
E.1 Suggested Semester Work Plan . . . 47
E.2 Review 1: Topic Approval and Feasibility . . . 47
E.3 Review 2: Midterm Review . . . 47
E.4 Review 3: Final Presentation and Viva . . . 48
E.5 Fortnightly Progress Log Template . . . 48

F Additional Discussion Notes **49**
F.1 Why Permissioned Blockchain? . . . 49
F.2 Why Separate Prescription and Drug Contracts? . . . 49

F.3 Why Keep Medical Records Off-Chain? 49
F.4 Atomicity as a Safety Property 50
F.5 Interpretation of the 100% Security Result 50
F.6 Interpretation of the Storage Result 50
F.7 Engineering Trade-offs 50

List of Figures

3.1 High-level MedTrust system architecture 13
3.2 MedTrust point-of-sale verification workflow 17

4.1 Total end-to-end verification latency across the eight workloads 22
4.2 Illustrative reproduction of the throughput trend reported in the research paper 24
4.3 Ledger footprint trend based on the reported per-transaction storage figures 25

A.1 Prescription registration flowchart 40
A.2 Drug provenance registration flowchart 41
A.3 Atomic dispensing decision flowchart 41

List of Tables

2.1 Comparative view of literature paradigms 9
2.2 Detailed literature review matrix 10

3.1 Software and infrastructure components 19

4.1 Eight pharmaceutical workload datasets used in evaluation 21
4.2 End-to-end latency decomposition across workloads 22
4.3 Reported throughput comparison 23
4.4 Ledger storage comparison at 100,000 transactions 24
4.5 Reported security interception outcomes 25
4.6 Reported comparison with selected architectural paradigms 26

C.1 Consolidated experimental workload observations 44
C.2 Latency observations reproduced from the supplied research paper . . . 44
C.3 Security interception observations 45

D.1 Experimental testbed specification 46
D.2 Cryptographic components used in the proposed design 46

E.1 MedTrust project development schedule 47
E.2 Fortnightly progress log template 48

Abbreviations

ABDM	Ayushman Bharat Digital Mission
API	Application Programming Interface
CA	Certificate Authority
CDSCO	Central Drugs Standard Control Organisation
CFT	Crash Fault Tolerant
CSP	Cryptographic Service Provider
DPDP	Digital Personal Data Protection
ECDSA	Elliptic Curve Digital Signature Algorithm
EHR	Electronic Health Record
EUF-CMA	Existential Unforgeability under Chosen Message Attack
FHIR	Fast Healthcare Interoperability Resources
GS1	Global Standards 1
HIS	Hospital Information System
IPFS	InterPlanetary File System
MSP	Membership Service Provider
NDC	National Drug Code
NIST	National Institute of Standards and Technology
P-256	NIST prime-field elliptic curve with 256-bit parameters
PDMP	Prescription Drug Monitoring Program
PII	Personally Identifiable Information
PKI	Public Key Infrastructure
POS	Point of Sale
PPT	Probabilistic Polynomial-Time
QR	Quick Response
Raft	Crash Fault Tolerant consensus protocol
Rx	Prescription
TPS	Transactions Per Second
ZKP	Zero-Knowledge Proof

Chapter 1

Introduction

1.1 Introduction to the Project

The capstone project is intended to consolidate engineering knowledge through a practical design experience involving problem identification, literature review, system design, implementation, testing, validation, documentation, and communication. The supplied institutional guidelines specifically require a project to begin with a clearly defined problem, identify research gaps, formulate a feasible design, implement and validate a prototype or proof of concept, and summarize the findings in a technical report.

MedTrust addresses a problem at the intersection of healthcare information systems, pharmaceutical supply-chain integrity, cryptography, and permissioned blockchain technology. The project focuses on the final dispensing event because this is the point at which two independent assertions must become true at the same time: the prescription must be legitimate and unfulfilled, and the physical drug package must be authentic, within its usable period, not recalled, and under the expected custody of the dispensing pharmacy.

The research paper describes this as a *dual-layer verification gap*. Existing systems can protect one side of the transaction while leaving the other side outside the same authorization boundary. MedTrust therefore treats prescription authenticity and drug provenance as two synchronized verification layers rather than independent checks.

1.2 Background of the Project Topic

Pharmaceutical distribution involves multiple organizational boundaries. A medicine can pass from a manufacturer to a distributor and finally to a pharmacy, while a medical authorization is created independently by a licensed practitioner. Digital prescription infrastructure can establish that a prescription was issued by an authenticated practitioner, but this does not necessarily establish that the physical package presented at the pharmacy corresponds to a valid supply-chain record. Likewise, a traceability platform can record package custody without establishing that the person attempting to redeem the package

has an authentic, active prescription.

The supplied research work identifies two principal operational risks. The first is prescription duplication and diversion, in which an electronic authorization or credential is replayed across multiple dispensing locations. The second is supply infiltration, in which a legitimate prescription is used to obtain a counterfeit, expired, recalled, or otherwise unsuitable medicine. These risks are not necessarily caused by failure of modern cryptographic primitives; instead, they arise when two separately verified objects are not bound into one dispensing decision.

The proposed architecture uses a permissioned ledger because the participating entities are known organizations with different operational responsibilities. Hyperledger Fabric 2.5 provides an enterprise-oriented permissioned ledger model, while the research implementation uses Raft consensus, X.509 identities, CouchDB world state, and Go chaincode.

1.3 Motivation

The motivation for MedTrust is to establish a single, auditable transaction boundary for a dispensing event. The system should answer the following questions before allowing a medicine to be dispensed:

1. Is the prescription associated with an authenticated prescriber?
2. Is the prescription cryptographically intact?
3. Is the prescription still active and within its validity window?
4. Has the prescription already been fulfilled?
5. Is the physical drug package registered as an authentic asset?
6. Is the package still within its expiration boundary?
7. Is the package subject to an active regulatory recall?
8. Does the recorded supply-chain custodian match the dispensing pharmacy?
9. Can the verification be performed without placing patient PII on the ledger?
10. Can all of these conditions be evaluated as one atomic transaction?

These questions translate the broad safety objective into deterministic engineering requirements.

1.4 Scope of the Project

The scope of the project is the design and evaluation of a permissioned blockchain framework for pharmaceutical dispensing. The scope includes:

- prescription lifecycle management;
- cryptographic prescription verification;
- serialized drug provenance verification;
- custody validation;
- expiration and recall validation;
- atomic transaction coordination;
- zero-PII on-chain data design;
- protected off-chain storage for sensitive information;
- benchmarking of latency and throughput;
- security interception testing across defined adversarial cases; and
- comparison with selected architectural paradigms.

The supplied paper does not present a deployed national healthcare production system. The reported experiments are simulated dispensing cycles on a controlled benchmarking testbed. Accordingly, this report treats the measured values as experimental results of the prototype evaluation rather than as field measurements of an operational national network.

1.5 Problem Statement

Existing pharmaceutical verification architectures frequently separate clinical authorization from physical drug provenance. A dispensing system may therefore accept a valid prescription while being unable to establish the provenance of the medicine, or may establish package provenance while being unable to verify that the prescription is authentic, active, and unspent.

The problem addressed by MedTrust can be stated as follows:

> *Design a permissioned, privacy-preserving verification framework that synchronously validates the authenticity and lifecycle of a digital prescription together with the provenance, custody, expiration, and recall status of the physical drug package, and commits the dispensing state only when all required assertions succeed.*

1.6 Formal Statement of the Verification Gap

Let a dispensing transaction be represented as

$$T = \langle P, D, U_{\text{pharm}}, \tau_{\text{disp}} \rangle, \tag{1.1}$$

where P denotes the digital prescription, D denotes the physical drug package, U_{pharm} is the authenticated dispensing terminal, and τ_{disp} is the dispensing timestamp.

A fragmented architecture can be abstracted using two predicates:

$$F_{\text{clinical}}(P) \in \{0, 1\}, \qquad F_{\text{logistics}}(D, U_{\text{pharm}}) \in \{0, 1\}. \tag{1.2}$$

MedTrust replaces independent authorization with the conjunction

$$\Omega_{\text{MedTrust}}(T) = F_{\text{clinical}}(P) \wedge F_{\text{logistics}}(D, U_{\text{pharm}}) \wedge AtomicCommit(P, D). \tag{1.3}$$

Thus, the ledger transition is permitted only when the prescription and physical package satisfy their respective constraints within the same transactional boundary.

1.7 Objectives

The project objectives are:

1. To study existing blockchain-based e-prescription and pharmaceutical provenance systems.
2. To identify the architectural gap between clinical authorization and physical drug validation.
3. To design a permissioned blockchain architecture based on Hyperledger Fabric 2.5.
4. To design a cryptographic zero-PII prescription representation using salted SHA-256 and ECDSA.
5. To implement separate prescription and drug provenance verification layers.
6. To coordinate the layers using an Atomic Gateway Contract.
7. To evaluate latency, throughput, storage footprint, and security interception.
8. To analyze privacy and regulatory considerations relevant to the proposed data model.
9. To document limitations and define realistic future extensions.

1.8 Salient Contributions

The project research identifies four central contributions:

1. **Dual-layer conjunction:** clinical and physical verification are evaluated together.
2. **Privacy-preserving ledger representation:** cleartext patient identifiers and medical history are excluded from the distributed ledger.
3. **Atomic coordination:** a gateway contract commits both state transitions or commits neither.
4. **Empirical evaluation:** the prototype is evaluated using 8 therapeutic workload categories and 5,230 simulated dispensing cycles.

1.9 Organization of the Report

Chapter 1 introduces the project, motivation, scope, problem statement, and objectives. Chapter 2 presents the literature survey and research gap. Chapter 3 describes the methodology, system architecture, cryptographic protocol, smart contracts, implementation components, and workflow. Chapter 4 presents the experimental setup, results, and analysis. Chapter 5 discusses advantages, limitations, applications, security and privacy considerations. Chapter 6 provides the conclusion and future scope. Appendices contain representative algorithms, data structures, experimental records, and project documentation.

Chapter Summary

This chapter established the problem addressed by MedTrust and translated it into a formal dual-layer verification requirement. The next chapter reviews prior work across e-prescriptions, pharmaceutical traceability, healthcare privacy, and centralized prescription monitoring to identify the research gap that motivates the proposed architecture.

Chapter 2

Literature Survey

2.1 Introduction

The literature relevant to MedTrust can be organized into four broad areas: distributed e-prescription architectures, blockchain-based pharmaceutical traceability, decentralized healthcare privacy, and centralized prescription monitoring. The supplied research paper uses these areas to show that previous systems generally optimize one side of the dispensing problem rather than synchronously binding clinical authorization to physical drug provenance.

The literature survey follows the capstone guideline that the review should begin with the overall topic, discuss relevant research comparatively, and use the resulting observations to identify a research gap and define the project problem.

2.2 Distributed E-Prescription Architectures

Blockchain-based electronic prescription systems have been proposed to improve authorization, traceability, and patient control. The CEUR e-prescription approach in [1] combines decentralized storage and execution mechanisms, but the research paper notes that its cryptographic boundary terminates at digital authorization. It does not bind the prescription to the physical medicine dispensed at the counter.

The architecture described in [2] combines two blockchains with a digital wallet to support electronic prescription exchange. This direction is relevant to patient-held credentials and distributed authorization, but it still does not create a single atomic state transition linking prescription validity to a serialized drug package.

Alalwan et al. [3] studied attitudes of patients, pharmacists, and prescribers toward blockchain and machine learning in e-prescription systems. Their work is relevant to adoption considerations because technical correctness alone does not establish operational acceptance. For MedTrust, this supports the need to keep the dispensing workflow explicit and deterministic rather than making the pharmacist responsible for interpreting an opaque risk score.

The shared virtual ledger approach described in [4] considers medical fraud and misuse using a ledger-based model. Such approaches demonstrate the value of shared transaction histories, but the research gap remains when physical pharmaceutical inventory is not part of the same verification boundary.

The optimized blockchain-based medical prescription-management system in [5] addresses security and prescription management. Its relevance to MedTrust lies in showing how blockchain can be applied to authorization state, while the MedTrust contribution extends the state model to include a physical drug asset.

2.3 Blockchain-Based Pharmaceutical Traceability

Pharmaceutical traceability research focuses on preventing counterfeit drugs and preserving custody records. MedLedger [6] uses Hyperledger Fabric for drug traceability. This is architecturally close to MedTrust because both use permissioned blockchain infrastructure, but MedLedger's primary objective is physical drug provenance rather than synchronous prescription authorization.

PharmaChain [7] similarly concentrates on supply-chain provenance. The supplied research paper identifies the structural limitation as the absence of clinical prescription oversight. The system can establish a custody trail without proving that the patient-facing authorization is valid and unspent.

Kumar et al. [8] investigate blockchain strategies for counterfeit medication detection. Such work supports the use of immutable asset records and shared verification, but counterfeit detection alone does not resolve prescription replay.

The work in [9] discusses blockchain in pharmaceutical supply-chain management, emphasizing the role of distributed ledgers in increasing transparency. Reference [10] combines blockchain and big data integration for pharmaceutical supply-chain security. These studies reinforce the value of distributed provenance while leaving the clinical authorization layer as a separate concern.

Musamih et al. [12] present a blockchain approach to drug traceability in healthcare supply chains. Their work provides a strong foundation for the drug provenance side of MedTrust. MedTrust extends this concept by making the prescription state and drug state jointly necessary for dispensing.

Humayun et al. [19] investigate blockchain-based protection of drug distribution against tampering. The work is relevant to custody and integrity, while MedTrust additionally introduces the notion of a synchronized dispensing decision.

Saxena et al. [27] discuss blockchain for counterfeit drugs in the pharmaceutical industry. This supports the importance of immutable supply-chain events, while the MedTrust design focuses on the point at which the final package is released to the patient.

2.4 Healthcare Privacy and Zero-Knowledge Approaches

MedRec [11] is an early blockchain healthcare architecture that focuses on medical-data access and permission management. It demonstrates the value of patient-centered authorization but also illustrates why medical data should not simply be replicated in a ledger without a carefully defined privacy model.

Pancari et al. [13] compare Ethereum and Hyperledger Fabric in the context of access control. Their comparison is useful for understanding the distinction between public and permissioned ledger architectures. MedTrust adopts Hyperledger Fabric because the target participants are authenticated organizations rather than anonymous public users.

Al-Aswad et al. [14] propose a blockchain-based zero-knowledge model for healthcare security. This direction demonstrates that privacy-preserving cryptographic protocols can reduce disclosure. Sitaula et al. [24] examine zero-knowledge-proof-based privacy-preserving pharmaceutical traceability and recall. Chalkias et al. [25] develop policy-private blockchain authentication concepts. Li et al. [26] provide a systematic review of privacy preservation in blockchain-based healthcare data sharing.

These works motivate MedTrust's zero-PII architecture. The supplied research paper, however, uses a simpler operational model based on salted SHA-256 digests, ECDSA signatures, and encrypted off-chain storage rather than requiring zero-knowledge proofs for every point-of-sale transaction.

2.5 Centralized Prescription Monitoring

Prescription Drug Monitoring Programs (PDMPs) use centralized databases to track prescribing and dispensing. Wilson et al. [17], Rhodes et al. [22], and Smith et al. [18] study the effectiveness and consequences of PDMP use. These systems address prescription diversion and controlled-substance monitoring, but their centralized design creates a different trust and availability model from a permissioned consortium ledger.

The research paper identifies three structural concerns with centralized registries: concentrated data risk, inter-jurisdictional siloing, and lack of physical drug inspection. MedTrust does not claim that blockchain automatically solves every PDMP limitation; rather, it introduces a multi-organization verification layer in which the dispensing event can incorporate both prescription and physical-package state.

2.6 Systematic Comparative Review

Table 2.1 summarizes the architectural characteristics emphasized by the research paper.

Table 2.1: Comparative view of literature paradigms

Paradigm	**Prescription**	**Physical Provenance**	**Privacy Model**	**Main Gap**
E-prescription systems [1–5]	Strong focus	Limited	Varies	Physical package not atomically bound
Pharmaceutical ledgers [6–10,12,16,19,27]	Limited/absent	Strong focus	Varies	Clinical authorization not bound
Healthcare blockchain privacy [11,13–15,24–26]	Indirect	Indirect	Strong focus	Point-of-sale dual verification not central
PDMPs [17,18,22]	Strong focus	Absent	Centralized	Physical drug provenance absent
MedTrust	Enforced	Enforced	Zero-PII on-chain	Addresses dual-layer seam

2.7 Research Gap Identification

The literature reveals four recurring architectural patterns.

First, prescription-focused systems primarily answer *who authorized the medicine and whether the authorization is valid*. Second, traceability systems primarily answer *where the physical package originated and which organization currently possesses it*. Third, privacy systems answer *how sensitive medical information can be protected*. Fourth, centralized monitoring systems answer *how prescription behavior can be observed across dispensing events*.

MedTrust combines these concerns at the point of dispensing. The gap is not simply a missing feature; it is a missing transaction relationship. The project treats the prescription and the physical package as two claims that must be simultaneously true.

2.8 Research Gap to Problem Definition

The identified gap can be summarized as follows:

> Existing approaches commonly verify either the clinical authorization or the physical pharmaceutical asset. There is a need for a permissioned architecture in which both are evaluated synchronously and the final dispensing state is

committed atomically, while sensitive patient information remains off the shared ledger.

This gap directly leads to the MedTrust design presented in Chapter 3.

2.9 Literature Review Matrix

Table 2.2: Detailed literature review matrix

Ref.	Work	Primary focus	Relevance to MedTrus
[1]	Blockchain e-prescription security	Digital authorization and traceability	Establishes prescript side foundation
[2]	Two-blockchain prescription exchange	Wallet and prescription exchange	Motivates distributed dentials
[3]	Alalwan et al.	Stakeholder attitudes	Adoption and usability text
[4]	Shared virtual ledger	Medical fraud and misuse	Shared fraud records
[5]	Optimized prescription management	Secure prescription management	Prescription state man ment
[6]	MedLedger	Fabric drug traceability	Physical provenance f dation
[7]	PharmaChain	Drug supply-chain provenance	Custody tracking
[8]	Kumar et al.	Counterfeit detection	Anti-counterfeit contex
[9]	Pharmaceutical blockchain review	Supply-chain transparency	Industry context
[10]	Blockchain + big data	Supply-chain security	Data integration contex
[11]	MedRec	Medical data access	Patient privacy context
[12]	Musamih et al.	Drug traceability	Provenance architecture
[13]	Pancari et al.	Ethereum vs Fabric	Platform selection cont
[14]	BZKP	Healthcare privacy	Privacy-preserving des
[15]	Zakari et al.	Pharmaceutical blockchain review	Systematic industry con
[16]	Counterfeit medicine detection	Blockchain-based detection	Counterfeit mitigation
[17]	Wilson et al.	PDMP effectiveness	Centralized monitoring text
[18]	Smith et al.	PDMP effects	Monitoring trade-offs
[19]	Humayun et al.	Drug tampering	Distribution integrity
[20]	Hasnain et al.	Fabric for EHR security	Permissioned healthc ledger

Ref.	Work	Primary focus	Relevance to MedTrus
[21]	Surjandy et al.	Electronic drug prescription	Sociotechnical context
[22]	Rhodes et al.	PDMP systematic review	Prescription monitoring
[23]	Al Amin et al.	Blockchain fraud prevention	Multi-signature fraud trol
[24]	Sitaula et al.	ZKP pharmaceutical traceability	Privacy-preserving reca
[25]	Chalkias et al.	Policy-private authentication	Privacy and policy con
[26]	Li et al.	Healthcare blockchain privacy review	Privacy synthesis
[27]	Saxena et al.	Counterfeit drugs	Pharmaceutical authenti

Chapter Summary

The literature survey demonstrates that prescription integrity, drug provenance, privacy, and monitoring have each been studied extensively, but the research paper identifies a specific architectural seam between these domains. MedTrust responds by placing both verification layers inside an atomic permissioned transaction.

Chapter 3

Methodology and Implementation

3.1 Introduction

The methodology translates the identified research gap into a system that can be implemented and measured. The design follows a layered architecture: authenticated consortium identities form the trust boundary; a prescription contract validates the clinical authorization; a drug provenance contract validates the physical package; and an Atomic Gateway Contract coordinates both decisions.

The project research paper specifies Hyperledger Fabric 2.5, X.509 membership, Go chaincodes, CouchDB world state, MongoDB off-chain storage, a Node.js verification gateway, and React-based pharmacy terminals. This chapter explains the relationship among these components and the sequence of a dispensing transaction.

3.2 Design Requirements

3.2.1 Functional Requirements

1. Register a prescription with an authenticated prescriber identity.
2. Generate a salted cryptographic prescription digest.
3. Verify the prescription signature.
4. Enforce prescription lifecycle states.
5. Register serialized drug assets and their provenance.
6. Track custody handoffs.
7. Validate expiration and recall status.
8. Verify that the dispensing terminal is the recorded custodian.
9. Execute prescription and drug checks within one atomic gateway.
10. Emit a breach event when a required assertion fails.

3.2.2 Non-Functional Requirements

- **Privacy:** patient PII must not be committed as cleartext ledger data.
- **Integrity:** prescription authorization must be cryptographically verifiable.
- **Consistency:** successful dispensing must update both relevant lifecycle states.
- **Auditability:** consortium participants must be able to inspect authorized ledger events.
- **Performance:** the verification process should remain suitable for point-of-sale use.
- **Scalability:** the architecture should support concurrent pharmacy terminals.
- **Governance:** only authenticated consortium organizations should participate in the network.

3.3 High-Level System Architecture

Figure 3.1 represents the architecture reconstructed from the research paper's stakeholder and component descriptions.

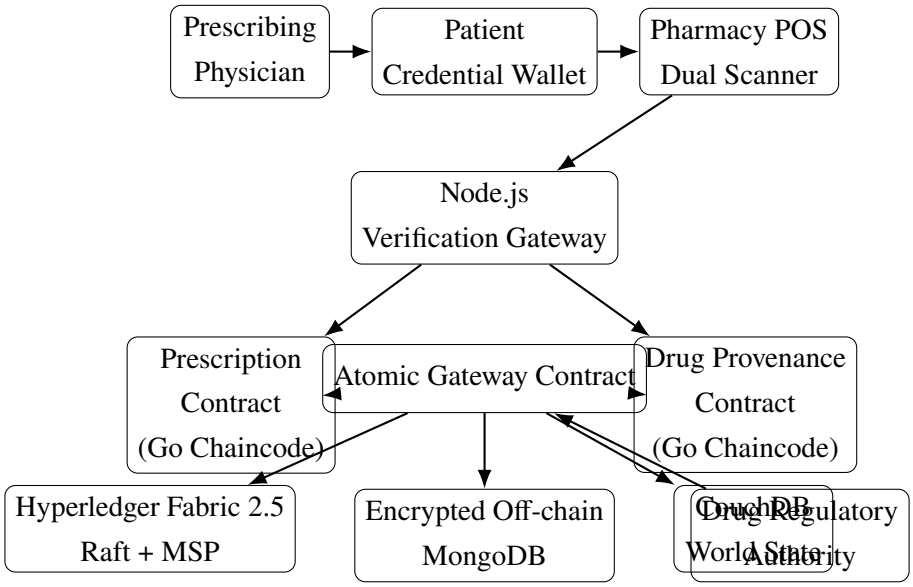


Figure 3.1: High-level MedTrust system architecture

3.4 Consortium Stakeholder Hierarchy

The research paper identifies six operational tiers.

3.4.1 Prescribing Physician

The physician organization is represented by `OrgPhysician`. An authenticated practitioner creates the clinical order, generates the salted prescription digest, and signs it using ECDSA. The ledger does not require the patient's cleartext clinical record.

3.4.2 Patient Credential Wallet

The patient wallet retains the prescription verification digest and transaction token required during dispensing. The research paper specifies that the wallet does not store cleartext diagnostic records as part of the proposed credential model.

3.4.3 Pharmaceutical Manufacturer

`OrgMfg` creates genesis consignment assets. The recorded package representation includes the DataMatrix identifier, drug identifier, production timestamp, and expiration boundary.

3.4.4 Logistics Distributor

`OrgDist` records custody handoffs and distribution checkpoints. This creates a sequence through which the system can verify whether the pharmacy claiming the package is the expected current custodian.

3.4.5 Retail Pharmacy

`OrgPharm` operates the point-of-sale terminal. The dual-scanning process captures the patient-side credential and the package-side DataMatrix identifier.

3.4.6 Drug Regulatory Authority

`OrgDRA` acts as the supervisory organization. It can inspect incident events and issue network-wide recall state changes for affected batches.

3.5 Identity and Membership

MedTrust is designed as a permissioned network. Participants are not anonymous public addresses; instead, the system uses X.509 certificates and Membership Service Providers. This arrangement supports organizational accountability and role-specific access.

The identity hierarchy can be represented as:

Root Governance → Certificate Authority → {Physician, Manufacturer, Distributor, Pharmacy, DRA} (3.1)

The patient wallet is treated as a credential holder rather than a consortium peer in the research architecture.

3.6 Privacy-Preserving Data Model

The privacy design is based on a simple principle: the ledger should contain enough information to verify integrity and state without reproducing the patient's medical record.

A 128-bit random salt is generated for the prescription:

$$S \leftarrow_R \{0,1\}^{128}. \tag{3.2}$$

The prescription digest is

$$H_{Rx} = SHA256(ID_{Patient} \| Lic_{Doc} \| DrugID \| Dosage \| S). \tag{3.3}$$

The physician signs the digest:

$$\sigma_{Doc} = ECDSA\text{-}Sign(d_{Doc}, H_{Rx}). \tag{3.4}$$

The proposed on-chain prescription state is:

$$S_{Rx} = \langle RxID, H_{Rx}, \sigma_{Doc}, Q_{Doc}, \tau_{issued}, \tau_{expires}, State \rangle. \tag{3.5}$$

The lifecycle state is one of

$$State \in \{Active, Fulfilled, Expired\}.$$

The cleartext medical information remains outside the shared ledger in encrypted storage.

3.7 Prescription Verification Module

The prescription contract performs three principal checks:

1. signature verification;
2. lifecycle verification; and
3. expiration verification.

The signature check can be expressed as

$$ECDSA\text{-}Verify(Q_{Doc}, H_{Rx}, \sigma_{Doc}) = 1. \tag{3.6}$$

The prescription must also satisfy

$$State = Active \quad \wedge \quad \tau_{disp} \leq \tau_{expires}. \tag{3.7}$$

If a prescription is already `Fulfilled`, replay redemption is rejected.

3.8 Drug Provenance Module

The drug provenance contract validates the physical package. The research paper specifies checks for package genesis, expiration, recall state, and custody.

The main conditions are:

$$Genesis(BatchID) = Valid, \tag{3.8}$$

$$\tau_{disp} \leq \tau_{drug_expires}, \tag{3.9}$$

$$isRecalled = 0, \tag{3.10}$$

$$CurrentCustodian(BatchID) = Org_{Pharm}. \tag{3.11}$$

These checks bind the dispensing decision to the package record rather than relying only on the patient's prescription.

3.9 Atomic Gateway Contract

The Atomic Gateway Contract is the central contribution of the architecture. It coordinates the two verification layers and commits state only after both succeed.

Let

$$C_{Rx} = F_{\text{clinical}}(P)$$

and

$$C_{Drug} = F_{\text{logistics}}(D, U_{\text{pharm}}).$$

The gateway accepts only when

$$C_{Rx} \wedge C_{Drug} = 1. \tag{3.12}$$

On success:

$$State_{Rx} \leftarrow Fulfilled, \qquad isDispensed_{Batch} \leftarrow 1. \tag{3.13}$$

On failure, the transaction is aborted and the research paper specifies that an immutable breach event is dispatched to the regulatory monitoring hub.

3.10 Transaction Workflow

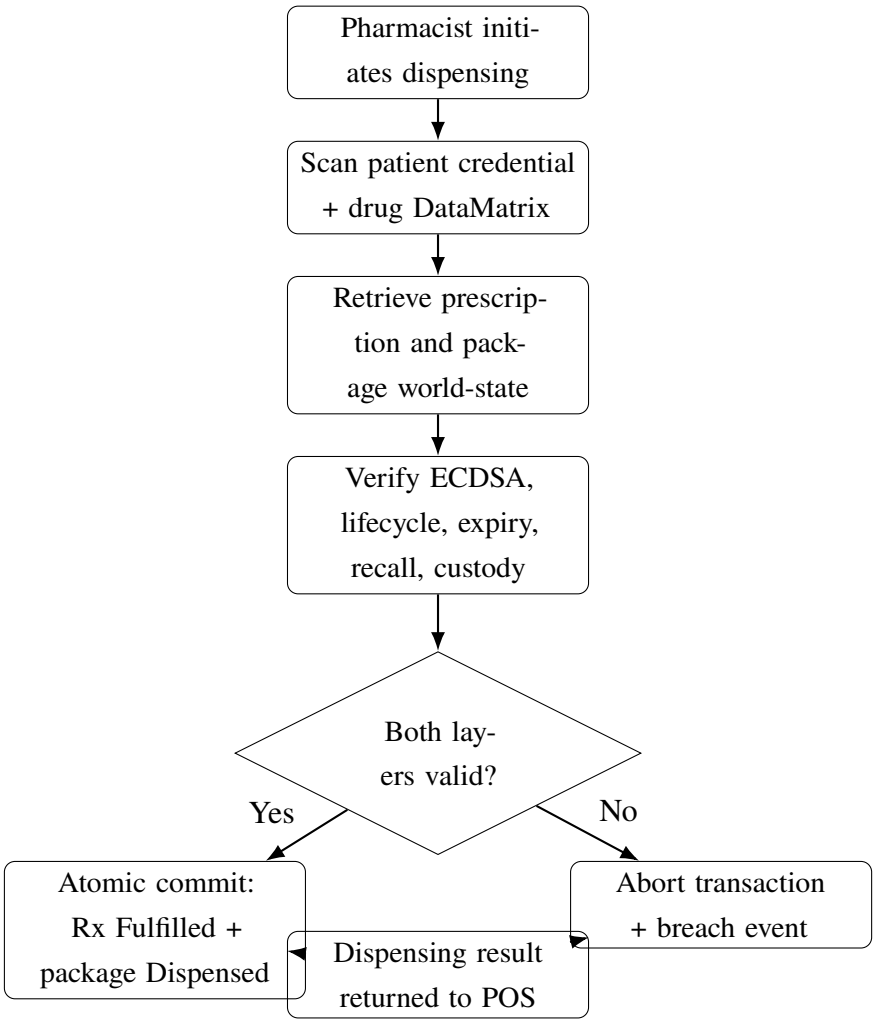


Figure 3.2: MedTrust point-of-sale verification workflow

3.11 Threat Model

The research paper models an active probabilistic polynomial-time adversary with six principal attack objectives.

3.11.1 Replay and Doctor-Shopping Fraud

An attacker may duplicate a valid patient credential and attempt redemption at multiple pharmacies. The prescription lifecycle lock prevents a successfully fulfilled prescription from being reused.

3.11.2 Counterfeit Packaging Injection

An attacker may create a package with a cloned or fabricated barcode. The drug provenance contract requires the package to correspond to a registered genesis asset.

3.11.3 Logistics Custody Hijacking

A genuine package may be diverted while in transit. The custody check compares the recorded current custodian with the dispensing organization.

3.11.4 Recall Evasion

A recalled lot must not be dispensed even when the prescription is valid. The drug state contains the recall flag checked by the provenance layer.

3.11.5 Expired Prescription Redemption

A prescription outside its validity window is rejected by the prescription contract.

3.11.6 Patient Re-identification

An observer may analyze immutable ledger events to infer medical information. The zero-PII design reduces direct exposure by storing salted digests rather than cleartext patient attributes.

3.12 Representative Chaincode Logic

The following listing is a compact representation of the decision logic described in the research paper. It is included for documentation and is not presented as a verbatim copy of a source-code repository.

Listing 3.1: Representative atomic gateway pseudocode

```
function verifyAndDispense(rxID, batchID, pharmacyID, timestamp):
    rx = getPrescription(rxID)
    drug = getDrug(batchID)

    rxOK = verifyECDSA(rx.publicKey, rx.digest, rx.signature)
```

```
            AND rx.state == "Active"
            AND timestamp <= rx.expiry

    drugOK = drug.genesisValid
              AND timestamp <= drug.expiry
              AND drug.isRecalled == false
              AND drug.currentCustodian == pharmacyID

    if rxOK AND drugOK:
        rx.state = "Fulfilled"
        drug.isDispensed = true
        commitAtomically(rx, drug)
        return "DISPENSED"
    else:
        emitBreachEvent(rxID, batchID)
        abortTransaction()
        return "REJECTED"
```

3.13 Implementation Stack

Table 3.1: Software and infrastructure components

Component	**Role in MedTrust**
Hyperledger Fabric 2.5	Permissioned distributed ledger platform
Raft	Crash Fault Tolerant consensus
Go chaincode	Prescription, provenance, and atomic gateway logic
CouchDB 3.3.2	World-state storage
MongoDB 7.0	Encrypted off-chain sensitive records
Node.js / Express	Verification gateway
React 18	Pharmacy terminal interface
X.509 / MSP	Organizational authentication and membership
SHA-256	Prescription digest construction
ECDSA / secp256r1	Prescription authorization signature
DataMatrix	Serialized drug-package identifier
WSL2 / Ubuntu 22.04 LTS	Experimental software environment

3.14 Testbed Configuration

The supplied paper reports an Intel Core i7-13700H workstation with 14 physical cores and 20 threads, 32 GB LPDDR5 memory, PCIe 4.0 NVMe storage, and Ubuntu 22.04 LTS through WSL2. The Fabric network uses six organization peer containers, a three-node Raft consensus cluster, CouchDB state engines, an off-chain MongoDB v7.0 cluster, and a Node.js Express gateway serving React 18 pharmacy terminals.

3.15 Methodological Sequence

The project methodology can be summarized into the following phases:

1. **Problem identification:** formalize the dual-layer verification gap.
2. **Literature review:** study e-prescription, provenance, privacy, and monitoring systems.
3. **Threat modeling:** define the six attack vectors.
4. **Architecture design:** select a permissioned consortium model.
5. **Cryptographic design:** specify salted hashing and ECDSA authorization.
6. **Smart-contract design:** separate clinical and physical verification.
7. **Atomic coordination:** enforce conjunction at the gateway.
8. **Benchmark design:** generate eight therapeutic workload categories.
9. **Evaluation:** measure latency, throughput, storage, and interception.
10. **Analysis:** interpret results against the stated requirements.

Chapter Summary

This chapter translated the research gap into a concrete architecture. The central design principle is that prescription state and drug state are independently verified but jointly committed. The next chapter evaluates the design through the workload and benchmark data reported in the supplied research paper.

Chapter 4

Results and Analysis

4.1 Introduction

The research paper evaluates MedTrust using eight standardized pharmaceutical workloads totaling 5,230 simulated dispensing cycles. The evaluation covers clean dispensing transactions and adversarial attempts across different therapeutic categories. This chapter reproduces the reported experimental values and analyzes them in terms of latency, concurrency, storage, security interception, and comparative architecture.

The values in this chapter are taken from the supplied research paper. They represent controlled prototype evaluation results rather than field measurements from deployed pharmacies.

4.2 Workload Design

Table 4.1: Eight pharmaceutical workload datasets used in evaluation

ID	Therapeutic class	Cycles	Clean	Attack	Threat vector
D1	Anti-infectives	650	400	250	Counterfeit batch injection
D2	Opioids and controlled	700	320	380	Replay fraud / doctor shopping
D3	Cardiovascular	800	620	180	High-throughput retail traffic
D4	Antidiabetics and insulin	750	550	200	Broken cold-chain custody
D5	Oncology and biologics	500	310	190	Regulatory recalls / sub-potency
D6	Respiratory and inhalers	600	460	140	Counterfeit aerosol packaging
D7	Gastrointestinal	650	510	140	Expired batch fulfillment
D8	Neuro-psychiatric	580	340	240	Prescription reuse / alteration
Total		**5,230**	**3,510**	**1,720**	**Holistic threat boundary**

The dataset contains 3,510 clean dispensing transactions and 1,720 adversarial attempts. Therefore, the reported evaluation distribution is

$$\frac{3510}{5230} \times 100 \approx 67.1\% \tag{4.1}$$

for clean transactions and

$$\frac{1720}{5230} \times 100 \approx 32.9\% \tag{4.2}$$

for adversarial attempts.

4.3 Latency Decomposition

Four timing components are measured:

1. world-state query;
2. cryptographic engine;
3. policy evaluation; and
4. atomic state commit.

Table 4.2: End-to-end latency decomposition across workloads

Dataset	Query	Crypto	Policy	Commit	Total
D1	0.42	1.18	0.18	0.65	2.43
D2	0.45	1.22	0.19	0.68	2.54
D3	0.41	1.15	0.17	0.64	2.37
D4	0.44	1.19	0.18	0.67	2.48
D5	0.43	1.21	0.19	0.66	2.49
D6	0.41	1.16	0.18	0.63	2.38
D7	0.42	1.17	0.17	0.65	2.41
D8	0.46	1.24	0.19	0.69	2.58
Mean ± Std	0.43±0.02	1.19±0.03	0.18±0.01	0.66±0.02	2.44±0.05

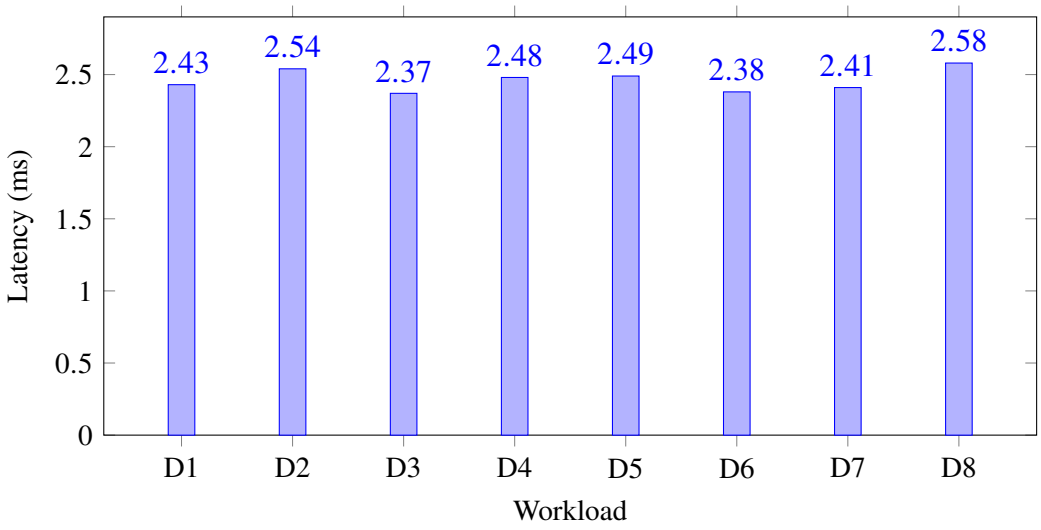


Figure 4.1: Total end-to-end verification latency across the eight workloads

The reported mean end-to-end latency is 2.44 ms with a standard deviation of 0.05 ms. The cryptographic engine is the largest component at 1.19 ms on average, accounting for

approximately 48.8% of total latency. The world-state query contributes 0.43 ms, policy evaluation 0.18 ms, and atomic commit 0.66 ms.

The small spread between the workload totals suggests that the measured latency is relatively stable within the controlled testbed. The highest reported workload latency is D8 at 2.58 ms, while D3 is 2.37 ms. The difference is 0.21 ms.

4.4 Concurrency and Throughput

The research paper evaluates concurrency from 10 to 500 simultaneous terminal threads. The reported peak is 1,690 transactions per second at 500 concurrent threads. The study also compares the system against an un-replicated PostgreSQL baseline, standard Hyperledger Fabric, and a public Ethereum baseline.

Table 4.3: Reported throughput comparison

System / configuration	Reported throughput
MedTrust at 500 terminals	1,690 TPS
Centralized PostgreSQL baseline	2,350 TPS
Ethereum baseline	approximately 15 TPS

The research paper reports that MedTrust reaches 71.9% of the raw throughput of the centralized PostgreSQL baseline:

$$\frac{1690}{2350} \times 100 \approx 71.9\%. \tag{4.3}$$

This comparison is useful because the centralized database is a different architecture. The result therefore should be interpreted as a throughput trade-off between replicated, authenticated consortium processing and an un-replicated relational baseline, not as a claim that the two systems provide identical security properties.

4.5 Throughput Plot

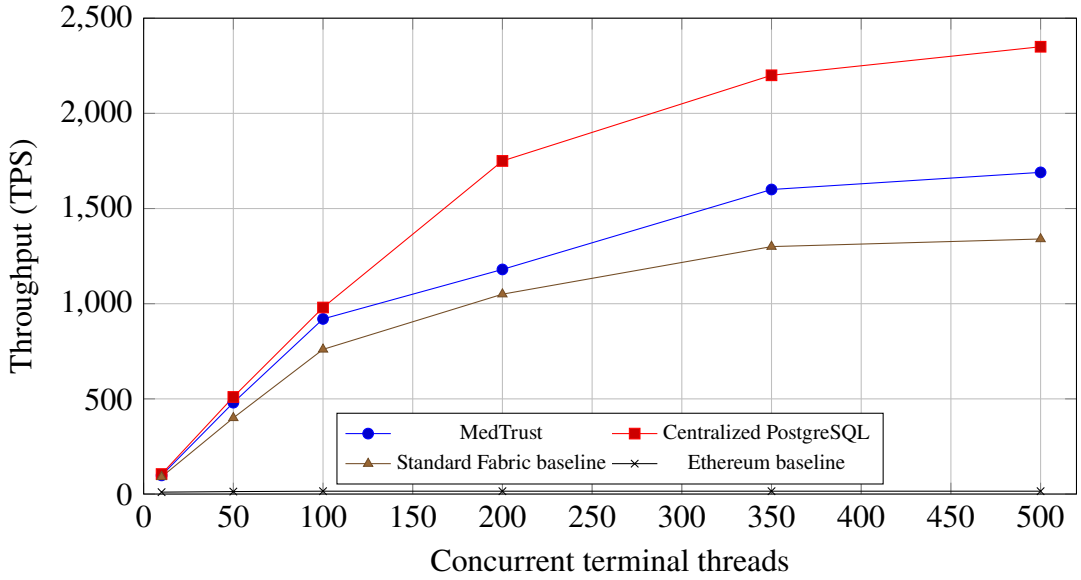


Figure 4.2: Illustrative reproduction of the throughput trend reported in the research paper

The plot uses the endpoint and trend values described in the research paper. Only the MedTrust peak, centralized PostgreSQL peak, standard Fabric comparative trend, and Ethereum ceiling are treated as reported comparison values; intermediate values are shown as a visual reconstruction of the paper's reported trend and should not be interpreted as additional independently measured observations.

4.6 Ledger Storage Optimization

The research paper compares three storage approaches. A monolithic healthcare blockchain is stated to require approximately 12.5 KB per transaction, Ethereum state storage approximately 18.2 KB, and the MedTrust representation 0.48 KB per transaction.

At 100,000 transactions, the reported footprints are:

Table 4.4: Ledger storage comparison at 100,000 transactions

Architecture	Storage	Relative observation
Monolithic EHR blockchain	1.22 GB	Baseline
Ethereum state storage	Approximately 1.82 GB	Highest among compared storage models
MedTrust	46.8 MB	96.2% reduction vs. monolithic baseline

The reduction is derived from the compact representation of cryptographic digests and lifecycle flags rather than storing complete clinical records on the ledger.

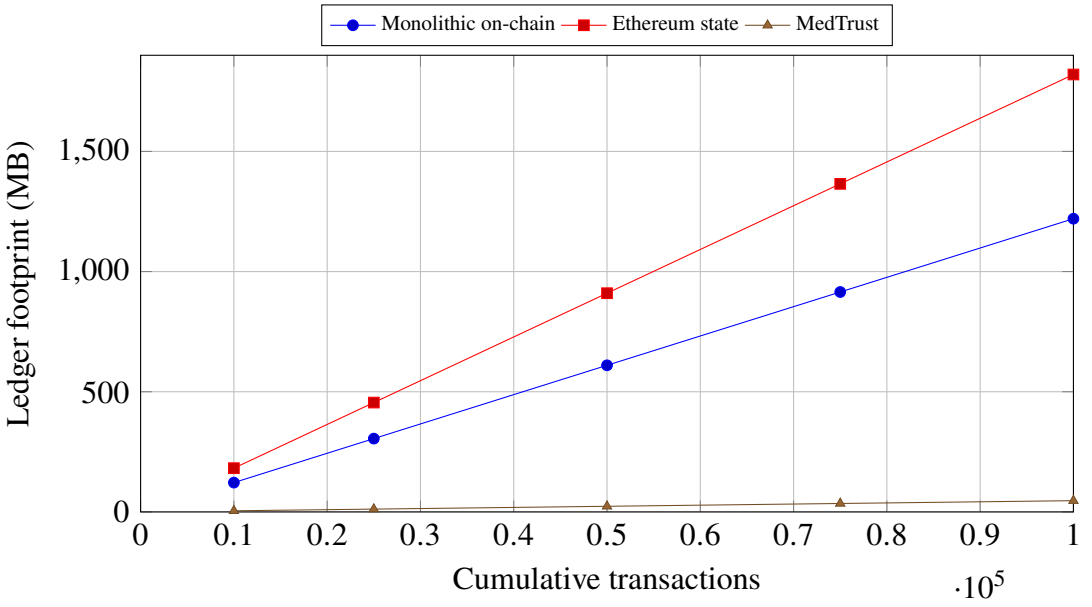


Figure 4.3: Ledger footprint trend based on the reported per-transaction storage figures

4.7 Security Interception

The research paper reports 3,510 legitimate transactions and 1,720 adversarial attempts. The adversarial categories include replay, counterfeit packaging, broken custody, recalled lots, and expired prescriptions.

Table 4.5: Reported security interception outcomes

Category	Attempts	Intercepted	Rate
Legitimate orders	3,510	3,510 authorized	100% correct authorization
Replay / diversion	460	460	100%
Counterfeit lot	380	380	100%
Broken custody	430	430	100%
Regulatory recall	240	240	100%
Expired prescription	210	210	100%

The paper reports an overall classification accuracy of 100.0%, false-positive rate of 0.00%, and F1-score of 1.000 for the deterministic test set.

These values should be interpreted in the context of the evaluation design: the test cases were generated to exercise deterministic rule boundaries. They do not establish that an operational national network will encounter zero errors under every possible real-world condition. Real-world performance would additionally depend on sensor reliability, data quality, key management, organizational integration, network availability, governance, and adversarial behaviors not represented in the simulated test set.

4.8 Cryptographic Security Analysis

The research paper states three principal guarantees.

4.8.1 Digest Integrity

The prescription digest is based on SHA-256 and incorporates a 128-bit salt. The salt increases the entropy of the input representation and makes precomputed dictionary matching substantially more difficult when the underlying patient attributes have predictable structure.

The report does not interpret the digest as reversible encryption. It is a one-way integrity representation. Sensitive cleartext remains off-chain.

4.8.2 Signature Unforgeability

ECDSA over the secp256r1/P-256 curve is used for physician authorization. Verification requires the corresponding public key and signed digest. An alteration to the signed message invalidates the signature verification.

4.8.3 Atomicity

Atomicity is the architectural property that distinguishes the MedTrust design. A prescription cannot transition to `Fulfilled` independently of the drug asset's successful validation. Likewise, the package cannot be marked as dispensed if the prescription layer fails.

4.9 Comparative Evaluation

Table 4.6: Reported comparison with selected architectural paradigms

Metric	Ethereum E-Rx	PharmaChain	Centralized PDMP	MedTrust
Dual-layer verification	Rx only	Drug only	Rx only	Synchronous
Transaction latency	≈15,000 ms	≈85 ms	≈1.8 ms	2.44 ms
Peak throughput	≈15 TPS	≈820 TPS	≈2,350 TPS	1,690 TPS
Transaction fee	$2.50–$24.00	$0.00	$0.00	$0.00
Patient PII on ledger	High/partial	N/A	Very high	None in proposed ledger model
Recall speed	Hours/days	Seconds	Minutes	Instantaneous state update in network
Consensus / architecture	Proof-of-Stake	Raft/CFT	RDBMS	Raft/CFT

The table should be interpreted as a system-level comparison reported by the research paper. The architectures differ in governance, trust assumptions, data models, and operational context, so a single metric cannot represent the full system behavior.

4.10 Interpretation of Results

The experimental evidence supports four observations within the tested environment.

First, the atomic gateway adds a relatively small measured latency overhead. The mean verification latency is 2.44 ms, with cryptographic processing being the largest timing component.

Second, the permissioned network sustains 1,690 TPS at 500 concurrent terminal threads in the reported benchmark. This is below the un-replicated PostgreSQL baseline but substantially above the public Ethereum comparison used in the study.

Third, the zero-PII ledger representation greatly reduces ledger growth compared with storing complete EHR/FHIR-style structures on-chain. The reported 46.8 MB footprint at 100,000 transactions is the direct consequence of the compact state representation.

Fourth, the deterministic rules intercepted all test attacks in the supplied workload. This result is expected for attack classes whose defining properties are explicitly represented as ledger state, such as a recalled flag, expired timestamp, invalid custodian, or fulfilled prescription.

4.11 Result Validation Considerations

The capstone guidelines require results to be thoroughly evaluated and logically connected to conclusions. For MedTrust, the following validation considerations are important:

- Repeatability of the same workload under identical testbed conditions.
- Variation in concurrency beyond the reported 500-thread ceiling.
- Multi-host deployment rather than a single workstation.
- Failure testing for peer, database, gateway, and network components.
- Evaluation with noisy or delayed custody updates.
- Testing of key rotation and certificate revocation.
- Validation of recall propagation under network partition scenarios.

These are not additional results claimed by the paper; they are recommended extensions of the evaluation methodology.

Chapter Summary

The reported evaluation demonstrates deterministic interception for the tested attack classes, sub-3-ms average verification latency, 1,690 TPS peak throughput, and substantial storage reduction. The results are encouraging within the controlled testbed, while the

analysis also identifies the difference between simulated deterministic workloads and future production-scale validation.

Chapter 5

Advantages, Limitations and Applications

5.1 Introduction

A capstone design should be evaluated not only by measured performance but also by its operational advantages, assumptions, limitations, and possible application domains. This chapter therefore separates the strengths demonstrated by the architecture from the constraints that must be addressed before real-world deployment.

5.2 Advantages

5.2.1 Synchronous Dual-Layer Verification

The principal architectural advantage is that clinical authorization and physical drug provenance are evaluated as one dispensing condition. This prevents a successful result from one layer from automatically authorizing the transaction when the other layer fails.

5.2.2 Atomic State Transition

The Atomic Gateway Contract ensures that the prescription and drug states move together. A successful dispensing event updates both the prescription state and the package dispensing state. A failed condition leaves the ledger state unchanged.

5.2.3 Privacy-Aware Ledger Design

The proposed design avoids placing patient names, diagnostic classifications, or treatment regimens on the distributed ledger. The research paper instead stores salted cryptographic digests and lifecycle tokens while sensitive cleartext is kept off-chain.

5.2.4 Permissioned Governance

Authenticated consortium membership is suitable for a setting involving physicians, manufacturers, distributors, pharmacies, and regulators. X.509 certificates and MSP roles allow organizational identity to be represented explicitly.

5.2.5 Auditability

Ledger events provide a shared history of prescription and drug-state transitions. Regulatory participants can observe breach events and recall state updates.

5.2.6 Fast Point-of-Sale Verification

The reported mean latency of 2.44 ms indicates that the prototype's verification path is lightweight under the tested environment.

5.2.7 Scalability in the Reported Benchmark

The reported peak throughput of 1,690 TPS at 500 concurrent terminal threads indicates that the prototype can process a substantial number of concurrent verification requests in the controlled benchmark.

5.2.8 Storage Efficiency

The proposed compact on-chain state avoids the storage growth associated with placing full medical records or complex healthcare documents directly on-chain.

5.3 Limitations

5.3.1 Prototype-Level Evaluation

The experimental results are based on simulated dispensing cycles rather than a national production deployment. The reported 100% interception rate therefore describes the tested deterministic cases, not every possible real-world fraud pattern.

5.3.2 Single-Workstation Testbed

The reported benchmark uses a workstation running Ubuntu through WSL2. A distributed deployment across geographically separated hosts could introduce network latency, hardware variability, and additional failure modes.

5.3.3 Dependence on Data Quality

A blockchain cannot correct incorrect input data. If a legitimate package is registered incorrectly, or a custody event is not recorded, the ledger can preserve the incorrect state immutably.

5.3.4 Identity and Key Management

ECDSA security depends on secure private-key handling. A compromised physician key or organizational credential could undermine authorization until the credential is revoked.

5.3.5 Integration Complexity

Real deployment would require interoperability with existing hospital information systems, pharmacy software, manufacturer serialization infrastructure, logistics platforms, and regulatory systems.

5.3.6 Privacy Is Not Equivalent to Complete Anonymity

Although the proposed ledger stores no cleartext patient PII, metadata and transaction timing can still be sensitive. Operational controls, access policies, and off-chain data protection remain necessary.

5.3.7 Regulatory Interpretation

The research paper discusses alignment with India's DPDP Act 2023. Actual legal compliance depends on the final deployment architecture, organizational roles, applicable rules, contracts, retention policies, and regulatory interpretation. A technical architecture should therefore not be treated as a substitute for legal review.

5.4 Applications

5.4.1 Retail Pharmacy Dispensing

The most direct application is pharmacy point-of-sale verification. The pharmacist scans the patient credential and drug package and receives a deterministic result.

5.4.2 Hospital Pharmacy

Hospital pharmacies could use the same dual-layer verification for internal dispensing, especially where medicines move across hospital inventory and controlled dispensing workflows.

5.4.3 High-Risk or Controlled Medicines

The replay and prescription lifecycle controls described in the research paper are particularly relevant to medicines where unauthorized duplication is an important operational concern.

5.4.4 Cold-Chain and Specialized Medicines

The provenance model can represent custody checkpoints. The research workload includes antidiabetic and insulin products with broken cold-chain custody as a threat category, indicating how the framework could be extended with sensor-derived custody or environmental assertions.

5.4.5 Regulatory Recall Enforcement

The regulatory authority can broadcast a batch recall by changing the corresponding world-state flag. Pharmacy verification then rejects the recalled batch.

5.4.6 Pharmaceutical Distribution Networks

Manufacturers and distributors can use the ledger to establish serialized asset histories and custody transitions before the final dispensing event.

5.5 Ethical and Social Considerations

Healthcare systems involve sensitive information and therefore require a careful balance between auditability and privacy. The MedTrust design reduces direct patient exposure by avoiding cleartext PII on the shared ledger. However, ethical deployment would also require transparent governance, appropriate consent and access practices, secure identity management, and clearly defined responsibility for erroneous or fraudulent records.

The project also illustrates a broader engineering principle: immutable infrastructure should be used only for information that organizations are prepared to preserve. Sensitive medical records are therefore separated from immutable transaction metadata.

5.6 Security and Privacy by Design

Security is distributed across multiple controls:

- authenticated organization identities;
- ECDSA signatures for physician authorization;
- salted SHA-256 prescription digests;
- lifecycle locks against replay;
- serialized package genesis records;
- custody verification;
- expiration and recall checks;

- atomic commit behavior; and
- off-chain storage for sensitive cleartext.

No single mechanism is assumed to provide the entire security property. The security model is a composition of cryptographic integrity, authorization state, physical asset state, and transaction atomicity.

5.7 Operational Failure Scenarios

A robust deployment must also consider what happens when infrastructure is unavailable. Examples include a pharmacy losing network connectivity, a certificate authority being unavailable, a peer node failing, a recall arriving while a transaction is in progress, or an off-chain record being temporarily inaccessible.

The proposed architecture provides deterministic rules for ledger state, but the supplied paper does not provide a full disaster-recovery or offline-dispensing protocol. Such mechanisms should therefore be treated as future engineering work rather than as current capabilities.

Chapter Summary

MedTrust provides a clear architectural advantage by binding prescription and physical package verification. At the same time, its prototype evaluation, integration requirements, key-management dependencies, and governance requirements must be considered before real deployment. The next chapter consolidates the findings and defines future extensions.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This capstone project investigated the use of a permissioned blockchain to connect two verification layers that are commonly handled independently in pharmaceutical dispensing: clinical prescription authorization and physical drug provenance.

The research work formalizes the dispensing transaction as a combination of a digital prescription, a physical drug package, an authenticated pharmacy terminal, and a dispensing timestamp. The key design decision is that neither clinical validation nor logistics validation is sufficient alone. MedTrust therefore requires their conjunction within an atomic transaction.

The architecture is implemented conceptually around Hyperledger Fabric 2.5, with six organizational roles, X.509 membership, Raft consensus, CouchDB world state, encrypted off-chain MongoDB storage, Go chaincodes, a Node.js gateway, and React pharmacy terminals. The cryptographic protocol uses a 128-bit salt, SHA-256 digest construction, and ECDSA signatures over the P-256/secp256r1 curve.

The reported evaluation covers eight therapeutic workload categories and 5,230 simulated dispensing cycles. Within the tested deterministic threat classes, all reported attacks were intercepted, with a reported overall accuracy of 100.0%. The reported mean end-to-end latency was 2.44 ms, the peak throughput was 1,690 TPS under 500 concurrent terminal threads, and the reported ledger storage reduction was 96.2% compared with the monolithic on-chain baseline.

The principal contribution of the work is therefore architectural rather than merely cryptographic: the system defines a dispensing transaction in which the prescription and the physical package must both be valid before the world state can move to a completed dispensing state.

6.2 Future Scope

The research paper identifies several direct extensions.

6.2.1 Multi-Host Enterprise Testnet

The prototype can be moved from a controlled workstation to multiple physical or virtual hosts. This would enable evaluation of realistic network latency, peer failures, geographic distribution, and organization-specific infrastructure.

6.2.2 Fabric Gateway SDK Integration

Future versions can deploy compiled chaincodes through the Fabric Gateway SDK and replace prototype integration paths with a more complete enterprise client architecture.

6.2.3 Physical Barcode and RFID Hardware

The dual-scanner workflow can be connected to physical Bluetooth 2D barcode readers and RFID devices. This would permit evaluation of scanning reliability, device latency, package handling, and human interaction.

6.2.4 Interoperability with ABDM

The research paper identifies interoperability with India's Ayushman Bharat Digital Mission as a future direction. Such integration would require carefully defined identity, authorization, interoperability, and privacy interfaces.

6.2.5 Sensor-Assisted Cold-Chain Validation

The D4 workload models broken cold-chain custody. A future system could attach authenticated sensor events to package state so that environmental conditions become part of the provenance validation.

6.2.6 Key Rotation and Revocation

Production deployment requires lifecycle management for physician and organizational certificates, including revocation, rotation, emergency credential suspension, and recovery from key compromise.

6.2.7 Resilience and Disaster Recovery

Future evaluation should test peer failures, database failures, network partitions, gateway restarts, and recovery of incomplete or interrupted workflows.

6.2.8 Broader Adversarial Evaluation

The current evaluation focuses on deterministic categories represented in the workload design. Future research should include adversarial input generation, compromised endpoints, malicious insiders, metadata analysis, denial-of-service scenarios, and incorrect upstream data.

6.3 Final Remarks

MedTrust demonstrates how permissioned blockchain technology can be used as an integration mechanism rather than merely as a database replacement. The important engineering idea is to define the dispensing event as a multi-condition transaction whose final state cannot be reached unless the relevant clinical and physical assertions agree.

The project also demonstrates the importance of evaluating a system through multiple dimensions: security interception alone is insufficient, while throughput alone does not establish clinical correctness. Latency, storage, privacy, identity governance, auditability, interoperability, and failure handling must all be considered together.

References

[1] "A Blockchain Based Approach for E-Prescription System Security and Traceability," CEUR Workshop Proceedings, vol. 3616, 2024.

[2] "Design and implementation of a system for the exchange of electronic prescriptions based on the combination of two blockchains and a digital wallet application," Conference Publication, 2021.

[3] T. Alalwan et al., "Patients', pharmacists', and prescribers' attitude toward using blockchain and machine learning in a proposed ePrescription system: online survey," *JAMIA Open*, vol. 5, no. 1, ooab115, 2022.

[4] "Detecting medical fraud and medical misuse using a shared virtual ledger," U.S. Patent US10942956, 2021.

[5] "Towards an Optimized Blockchain-Based Secure Medical Prescription-Management System," *Future Internet (MDPI)*, vol. 16, no. 7, 243, 2024.

[6] "Blockchain Medledger: A Hyperledger Fabric Enabled Drug Traceability System for Counterfeit Drugs in Pharmaceutical Industry," *International Journal of Pharmaceutics*, vol. 597, 120235, 2021.

[7] "PharmaChain: Blockchain-based drug supply chain provenance verification system," *Heliyon*, vol. 9, no. 7, 2023.

[8] S. Kumar et al., "A research and enhancement strategy for detecting counterfeit medications utilizing blockchain technology," *IET Blockchain*, 2025.

[9] "Blockchain technology is revolutionizing the pharmaceutical industry's drug supply chain management," in *Proceedings of the 3rd International Conference on Computing Advancements*, ACM, 2025.

[10] "Enhancing Pharmaceutical Supply Chain Transparency and Security with Blockchain and Big Data Integration," *Procedia / ScienceDirect*, 2025.

[11] A. Azaria, A. Ekblaw, T. Vieira, and A. Lippman, "MedRec: Using Blockchain for

Medical Data Access and Permission Management," in *Proc. 2nd International Conference on Open and Big Data (OBD)*, IEEE, pp. 25–30, 2016.

[12] A. Musamih, K. Salah, R. Jayaraman, J. Arshad, M. Debe, Y. Al-Hammadi, and S. Ellaham, "A Blockchain-based Approach for Drug Traceability in Healthcare Supply Chain," *IEEE Access*, vol. 9, pp. 9728–9743, 2021.

[13] S. Pancari, A. Rashid, J. Zheng, S. Patel, Y. Wang, and J. Fu, "A Systematic Comparison between the Ethereum and Hyperledger Fabric Blockchain Platforms for Attribute-Based Access Control in Smart Home IoT Environments," *Sensors*, vol. 23, no. 19, 8319, 2023.

[14] H. Al-Aswad, W. M. El-Medany, C. Balakrishna, N. Ababneh, and K. Curran, "BZKP: Blockchain-based zero-knowledge proof model for enhancing healthcare security in Bahrain IoT smart cities and COVID-19 risk mitigation," *Arab Journal of Basic and Applied Sciences*, vol. 28, no. 1, pp. 154–171, 2021.

[15] N. Zakari, M. Al-Razgan, A. Alsaadi, H. Alshareef, H. Al Saigh, L. Alashaikh, M. Alharbi, R. Alomar, and S. Alotaibi, "Blockchain technology in the pharmaceutical industry: a systematic review," *PeerJ Computer Science*, vol. 8, e840, 2022.

[16] S. Zagare, M. Khodaskar, Y. Sonawane, and H. Verma, "Counterfeit Medicine Detection using Blockchain," *International Journal of Soft Computing and Engineering*, vol. 14, no. 2, 2024.

[17] M. N. Wilson, J. A. Hayden, E. Rhodes, A. Robinson, and M. Asbridge, "Effectiveness of Prescription Monitoring Programs in Reducing Opioid Prescribing, Dispensing, and Use Outcomes: A Systematic Review," *The Journal of Pain*, vol. 20, no. 12, pp. 1383–1393, 2019.

[18] N. Z. Y. Smith, J. D. Thornton, S. H. Fenton, D. Simmons, and T. Champagne-Langabeer, "Helpful, Unnecessary, or Harmful: A Systematic Review of the Effects of Prescription Drug Monitoring Program Use on Opioid Prescriptions," *Pharmacy*, vol. 10, no. 5, 119, 2022.

[19] M. Humayun, N. Z. Jhanjhi, M. Niazi, F. Amsaad, and I. Masood, "Securing Drug Distribution Systems from Tampering Using Blockchain," *Electronics*, vol. 11, no. 8, 1195, 2022.

[20] M. Hasnain, F. R. Albogamy, S. S. Alamri, I. Ghani, and B. Mehboob, "The Hyperledger fabric as a Blockchain framework preserves the security of electronic health records," *Frontiers in Public Health*, vol. 11, 1272787, 2023.

[21] Surjandy, C. Cassandra, and S. Rumangkit, "The Influence of Blockchain Technology on the Use of Electronic Drug Prescription Based on Cultural Safety Sociotechnical Framework," *International Journal of Safety and Security Engineering*, vol. 14, no. 4, pp. 1253–1259, 2024.

[22] E. Rhodes, M. Wilson, A. Robinson, J. A. Hayden, and M. Asbridge, "The effectiveness of prescription drug monitoring programs at reducing opioid-related harms and consequences: a systematic review," *BMC Health Services Research*, vol. 19, 784, 2019.

[23] M. Al Amin, R. Shah, H. Tummala, and I. Ray, "Utilizing Blockchain and Smart Contracts for Enhanced Fraud Prevention and Minimization in Health Insurance through Multi-Signature Claim Processing," arXiv preprint arXiv:2407.17765, 2024.

[24] A. Sitaula, M. A. Uddin, J. Ayoade, N. H. Chu, and R. Rafeh, "Zero-Knowledge Proof-Based Privacy-Preserving Pharmaceutical Traceability and Recall Using Blockchain," *Blockchains*, vol. 4, no. 1, 5, 2026.

[25] K. K. Chalkias, D. Maram, A. Roy, J. Wang, and A. Yadav, "Zero-knowledge Authenticator for Blockchain: Policy-private and Obliviously Updateable," IACR Cryptology ePrint Archive, 2024.

[26] K. Li, A. Lohachab, M. Dumontier, and V. Urovi, "Privacy preservation in blockchain-based healthcare data sharing: a systematic review," *Peer-to-Peer Networking and Applications*, vol. 18, 302, 2025.

[27] N. Saxena, I. Thomas, P. Gope, et al., "PharmaCrypt: blockchain for critical pharmaceutical industry to counterfeit drugs," *Computer*, vol. 53, no. 7, pp. 29–44, 2020.

Appendix A

Soft Code Flowcharts

A.1 Prescription Registration Flow

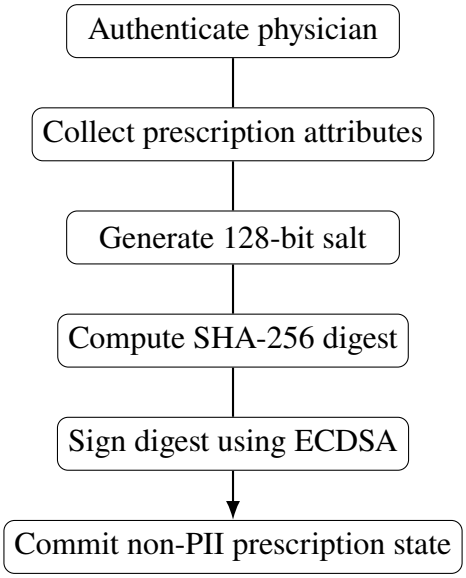


Figure A.1: Prescription registration flowchart

A.2 Drug Asset Registration Flow

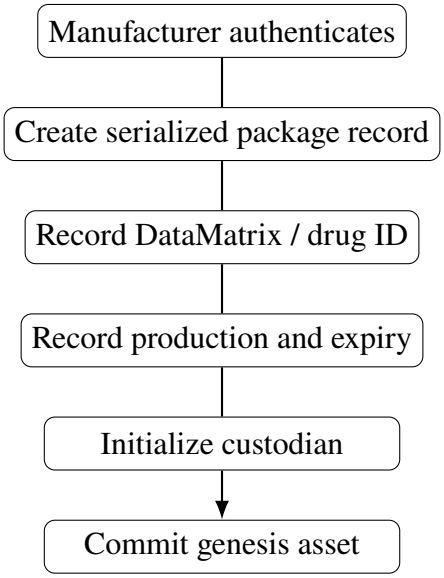


Figure A.2: Drug provenance registration flowchart

A.3 Atomic Dispensing Flow

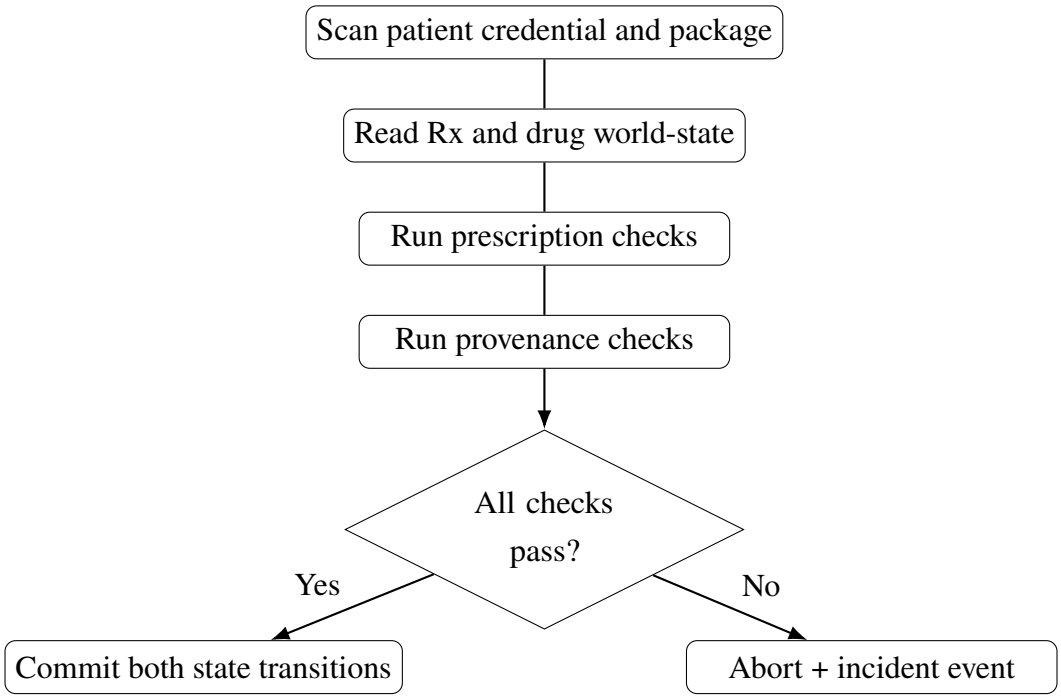


Figure A.3: Atomic dispensing decision flowchart

Appendix B

Data Structures and Sample Records

B.1 Prescription Ledger Record

Listing B.1: Representative non-PII prescription state

```
{
  "rxID": "RX-2026-000001",
  "hash": "<SHA-256 digest>",
  "signature": "<ECDSA signature>",
  "doctorPublicKey": "<X.509/public-key reference>",
  "issuedAt": "<timestamp>",
  "expiresAt": "<timestamp>",
  "state": "Active"
}
```

The record illustrates the data-minimization principle: the ledger state contains verification metadata rather than the patient's name, diagnosis, or treatment history.

B.2 Drug Asset Record

Listing B.2: Representative serialized drug state

```
{
  "batchID": "BATCH-000001",
  "drugID": "<registered drug identifier>",
  "dataMatrix": "<serialized package identifier>",
  "manufacturedAt": "<timestamp>",
  "expiresAt": "<timestamp>",
  "currentCustodian": "OrgPharm",
  "isRecalled": false,
  "isDispensed": false
}
```

B.3 Incident Event Record

Listing B.3: Representative breach event

```
{
  "eventType": "DISPENSING_REJECTED",
  "rxID": "<prescription identifier>",
  "batchID": "<batch identifier>",
  "reason": "<failed deterministic condition>",
  "timestamp": "<ledger timestamp>",
  "reportedTo": "OrgDRA"
}
```

Appendix C

Experimental Observations

C.1 Workload Totals

Table C.1: Consolidated experimental workload observations

Dataset	Total cycles	Clean	Adversarial
D1	650	400	250
D2	700	320	380
D3	800	620	180
D4	750	550	200
D5	500	310	190
D6	600	460	140
D7	650	510	140
D8	580	340	240
Total	5,230	3,510	1,720

C.2 Latency Observations

Table C.2: Latency observations reproduced from the supplied research paper

Dataset	Query	Crypto	Policy	Commit	Total
D1	0.42	1.18	0.18	0.65	2.43
D2	0.45	1.22	0.19	0.68	2.54
D3	0.41	1.15	0.17	0.64	2.37
D4	0.44	1.19	0.18	0.67	2.48
D5	0.43	1.21	0.19	0.66	2.49
D6	0.41	1.16	0.18	0.63	2.38
D7	0.42	1.17	0.17	0.65	2.41
D8	0.46	1.24	0.19	0.69	2.58

C.3 Security Interception Observations

Table C.3: Security interception observations

Attack category	Attempts	Intercepted
Replay / diversion	460	460
Counterfeit lot	380	380
Broken custody	430	430
Regulatory recall	240	240
Expired prescription	210	210
Total adversarial	1,720	1,720

Appendix D

System Components and Specifications

D.1 Computing Testbed

Table D.1: Experimental testbed specification

Processor	Intel Core i7-13700H
CPU configuration	14 physical cores / 20 threads
Peak clock	5.0 GHz
Memory	32 GB LPDDR5
Storage	PCIe 4.0 NVMe
Operating system	Ubuntu 22.04 LTS through WSL2
Ledger	Hyperledger Fabric 2.5
Consensus	Three-node Raft cluster
World state	CouchDB 3.3.2
Off-chain database	MongoDB 7.0
Gateway	Node.js Express
Client	React 18

D.2 Cryptographic Components

Table D.2: Cryptographic components used in the proposed design

SHA-256	One-way digest for prescription representation
128-bit salt	Randomization of digest input
ECDSA	Digital signature for physician authorization
secp256r1 / P-256	Elliptic curve used by ECDSA
X.509	Organizational identity certificate format
MSP	Fabric identity and membership management

Appendix E

Project Work Plan and Review Documentation

E.1 Suggested Semester Work Plan

The capstone guidelines require a project engagement schedule and development timeline. The following plan translates the MedTrust work into a semester-oriented schedule.

Table E.1: MedTrust project development schedule

Phase	Primary activity	Expected output
1	Topic finalization and feasibility	Approved problem statement
2	Literature survey	Research matrix and gap identification
3	Threat model	Attack-vector definition
4	Architecture design	System architecture and data model
5	Cryptographic design	Hash/signature protocol
6	Chaincode design	Prescription/provenance/gateway modules
7	Integration	Gateway and terminal workflow
8	Workload preparation	Eight therapeutic datasets
9	Performance testing	Latency and throughput observations
10	Security testing	Interception matrix
11	Comparative analysis	State-of-the-art comparison
12	Documentation	Interim/final report and presentation

E.2 Review 1: Topic Approval and Feasibility

The first review requires a concise project definition, objectives, scope, technologies, and a semester Gantt chart. For MedTrust, the review material should emphasize the dual-layer verification gap, permissioned blockchain feasibility, privacy model, and proposed evaluation methodology.

E.3 Review 2: Midterm Review

The second review requires literature understanding, problem clarity, system-level representation, implementation/results, and an interim report. This report provides the technical narrative needed for that evaluation, including the literature gap, architecture, algorithms, experimental setup, preliminary results, and future extensions.

E.4 Review 3: Final Presentation and Viva

The final review requires demonstration of working modules, testing, implementation details, technical documentation, outcomes, and conclusion. The project should demonstrate the prescription verification path, drug provenance path, atomic gateway behavior, and security rejection cases.

E.5 Fortnightly Progress Log Template

Table E.2: Fortnightly progress log template

Reporting period
Work completed
Technical issues
Evidence / result
Work planned next
Mentor comments
Mentor signature

Appendix F

Additional Discussion Notes

F.1 Why Permissioned Blockchain?

The proposed problem involves known organizations with defined responsibilities. A public, permissionless blockchain would introduce a different trust model and potentially unnecessary transaction-cost and privacy considerations. A permissioned network allows membership to be tied to organizational identity while retaining distributed state and multi-party verification.

The choice therefore follows the project requirements rather than assuming that blockchain is universally preferable to a conventional database. The research paper explicitly compares MedTrust against centralized PostgreSQL and public Ethereum baselines to illustrate the performance and governance trade-offs.

F.2 Why Separate Prescription and Drug Contracts?

Separation of the two contracts provides modularity. The prescription contract can evolve according to clinical authorization rules, while the drug provenance contract can evolve according to serialization, recall, and custody requirements. The Atomic Gateway then becomes the stable integration boundary.

This separation also helps audit failures. A rejected transaction can be associated with the verification layer whose condition failed.

F.3 Why Keep Medical Records Off-Chain?

Distributed ledgers are designed for shared state and immutability. Medical records, however, can contain highly sensitive and mutable information. Storing complete medical records on a shared immutable ledger can create unnecessary privacy and retention risks.

MedTrust therefore stores only a compact verification representation on-chain and keeps sensitive cleartext off-chain. This architecture also reduces storage requirements.

F.4 Atomicity as a Safety Property

Atomicity is not merely a database optimization in this project. It represents the safety condition that a successful dispensing state should correspond to simultaneous validity of the two independent claims. Without atomicity, an implementation could accidentally consume a prescription while rejecting the package, or mark a package as dispensed without a valid prescription.

F.5 Interpretation of the 100% Security Result

The reported 100% interception result is meaningful because the tested attack categories are deterministically represented in the state model. For example, a replay attack is represented by the prescription already being fulfilled; a recalled package has an explicit recall flag; and a custody violation is represented by a custodian mismatch.

However, the result should not be generalized beyond the tested cases. It does not measure attacks that are absent from the workload definition, nor does it measure human, organizational, or integration failures.

F.6 Interpretation of the Storage Result

The 96.2% storage reduction is primarily an architectural consequence of data minimization. The system does not attempt to make a complete electronic health record immutable. Instead, it preserves a compact proof-oriented representation and stores sensitive content separately.

This distinction is important when comparing blockchain systems: storage efficiency depends heavily on what each system chooses to put on-chain.

F.7 Engineering Trade-offs

MedTrust introduces several explicit trade-offs:

- distributed trust versus centralized simplicity;
- cryptographic verification versus computation;
- auditability versus metadata exposure;
- immutable state versus correction procedures;
- consortium governance versus open participation;
- compact on-chain state versus dependence on off-chain systems.

These trade-offs should be considered during future deployment studies.

Appendix Summary

The appendices provide flowcharts, representative data structures, experimental observations, hardware/software specifications, project scheduling documentation, and additional engineering discussion. They are intended to support the main report without replacing the core technical narrative.